Speciality: Computer Security (S7) **Course:** Advanced Networks

Exercise Session – Chapter 3: Quality of Service

Exercise 1:
Explain why QoS is necessary in modern networks. Give three real-world examples where QoS is critical and describe what happens if QoS is not implemented.

Exercise 2:

Define the following parameters and explain how each one affects user experience:

- Bandwidth
- Delay
- Jitter
- Packet loss
  Provide one example of an application sensitive to each.

Exercise 3:
Briefly describe the token bucket algorithm and its role in traffic shaping.
Then, suppose a bucket has a capacity of 10 tokens and a rate of 2 tokens/second. How long will it take to refill after a burst of 10 packets?

Exercise 4:

Given the following DSCP values, identify the type of traffic and priority:

1. 46
2. 0
3. 10
4. 34

Exercise 5:

Why do QoS mechanisms often depend on **Layer 3 and 4 information** (IP address, port number) rather than **Layer 2 (MAC address)**?

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Solution:

Exercise 1:

Why QoS is necessary:
Networks carry very different kinds of traffic (voice, video, bulk data) with different sensitivity to delay, jitter and loss. QoS gives the network the ability to *prioritize* and *control* traffic so that latency-sensitive and loss-sensitive applications keep working well even under congestion.

Three real-world examples and what happens without QoS:

1. VoIP / Phone calls (enterprise call system)
    - Requirement: very low latency and low jitter, low packet loss.
    - Without QoS: calls suffer from choppy audio, long delays (echo), dropped packets → poor call quality and user complaints.
2. Video conferencing (Zoom/Teams in a meeting)
    - Requirement: moderate to low latency, low jitter, reasonable bandwidth.
    - Without QoS: video freezes, lip-sync problems, dropped frames; interactive collaboration becomes difficult.
3. Real-time control / Telemedicine / Remote robot control
    - Requirement: ultra-low latency and reliability.
    - Without QoS: control commands are delayed or lost — can cause malfunction, safety risks, or loss of service.

(Other examples: online gaming — lag; financial trading — small delays cause big losses; cloud VoD streaming — rebuffering.)

Exercise 2:

1. Bandwidth (Throughput)
    - Definition: Maximum data rate a link or flow can carry (bps).
    - Effect if low: Applications stall or run slower (long downloads, poor video quality).
    - Example app sensitive to bandwidth: Video streaming (HD/4K).
2. Delay (Latency)
    - Definition: Time it takes a packet to travel from source to destination.
    - Effect if high: Noticeable lag in interactive apps.
    - Example app sensitive to delay: VoIP / remote control.
3. Jitter (Delay variation)

- Definition: Variability in packet inter-arrival time (variation of latency).
- Effect if high: Packets arrive irregularly causing choppy audio/video unless buffered.
- Example app sensitive to jitter: Real-time voice/video conferencing.

4. Packet loss
   - Definition: Percentage of packets that never reach their destination.
   - Effect if high: Retransmissions (for TCP), corrupted streams (for UDP) — poor user experience.
   - Example app sensitive to packet loss: VoIP and real-time video (UDP-based).

Exercise 3:

Short description (role in shaping/policing):

- The Token Bucket controls the transmission rate while allowing bursts. Tokens are added to a bucket at a steady rate R (tokens/sec). Each packet consumes tokens proportional to its size. If tokens are available, the packet is sent immediately (allows bursts up to the bucket size). If tokens are absent, the router either buffers the packet (shaping) or drops/remarks it (policing).

Numeric part (bucket = 10 tokens, rate = 2 tokens/sec):

- After sending a burst that consumes all 10 tokens, the bucket is empty.
- Time to refill to 10 tokens at 2 tokens/sec is:

time=10 tokens2 tokens/sec=5s

Exercise 4:

Given the following DSCP values, identify the **type of traffic** and **priority** (based on the DSCP table):
→ 46, 0, 10, 34

DSCP Value (Decimal)	**Binary**	**PHB (Per-Hop Behavior)**	**Traffic Type / Example**	**Priority Level-Drop Level**
46	101110	**Expedited Forwarding (EF)**	Real-time traffic (VoIP, video conferencing)	**Highest (low latency, low jitter)**
0	000000	**Best Effort (BE)**	Normal data (web browsing, email)	**Lowest**

10	001010	**Assured Forwarding (AF11)**	Low-priority assured traffic (file transfer, internal app data)	**Low–low**
34	100010	**Assured Forwarding (AF41)**	High-priority streaming or interactive video	**High-low**

Exercise 5: Why QoS mechanisms use Layer-3/Layer-4 info rather than Layer-2 (model answer)

Short answer: Because L3/L4 fields (IP addresses and TCP/UDP ports) carry *end-to-end and application* information needed to classify traffic across different networks, while L2 (MAC) addresses are only meaningful on a single local link.

Explanation (points you can use in class):

- **End-to-end visibility:** IP addresses and ports uniquely identify the communicating hosts and the application (e.g., destination IP and port 5060 → SIP). Routers on the path can use this to classify and enforce QoS consistently across multiple hops.
- **MACs are local:** A MAC address identifies a device only on the local LAN segment. Each router hop replaces the frame's MAC addresses — downstream routers cannot rely on an original MAC to identify the flow.
- **Application awareness:** Many QoS policies target *applications* (VoIP, HTTP, video). Ports (L4) and deep inspection of L3/L4 headers are the practical way to detect application flows.
- **Routing requirement:** QoS policies must follow routed paths; routers forward by IP. So QoS decisions tied to IP-based markers (DSCP) travel with the packet across networks, whereas MACs do not.
- **Exceptions & remarks:** In some controlled LAN environments, L2 (802.1p/CoS) is used for local switching priority, but it must be mapped to L3 DSCP on routers for end-to-end QoS. Also, encrypted traffic (tunneled or TLS) complicates L4 classification, which then requires alternative methods (e.g., marking at source, using metadata, or DSCP trust at edge).